

FDTD: diffraction over objects

Alichan Borokov Sam Hemati
Alichan.Borokov@UGent.be Sam.Hemati@UGent.be

Joran Van Haelst
Joran.VanHaelst@UGent.be

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1 Introduction

The objective is to simulate diffraction around an obstacle. To this end a FDTD simulation involving one source and three observers was made. The process starts simple with free space simulated with the help of Perfectly Matching layers or PML. Then a perfectly hard ground and perfectly hard infinitely thin wall is added of which the analytical solution exists. Then this concept is extended to a thick wall with a real impedance and lastly a wedge is studied through means of a triangular obstacle.

2 method

2.1 The equations

The equations to be solved are the following wave equation equations

$$\frac{\partial \mathbf{o}}{\partial t} + \nabla p = 0 \quad (1)$$

$$\frac{\partial p}{\partial t} + c^2 \nabla \cdot \mathbf{o} = 0 \quad (2)$$

or in cartesian coordinates

$$\frac{\partial o_x}{\partial t} + \frac{\partial p}{\partial x} = 0 \quad (3)$$

$$\frac{\partial o_y}{\partial t} + \frac{\partial p}{\partial y} = 0 \quad (4)$$

$$\frac{\partial p}{\partial t} + \frac{\partial o_x}{\partial x} + \frac{\partial o_y}{\partial y} = 0 \quad (5)$$

The fact that the simulated domain is limited means that there will be boundaries. This means there will always be reflections while the physical situation requires the waves to be able to leave the domain. To solve this problem, an absorbing layer can be included so that by the time the wave reaches the edge of the domain there is no wave left to reflect seeing as it is already fully absorbed. Unfortunately this creates it's own problems. The first one is that the sudden change of material from free space into an absorbing material will allow for both transmission and reflection. Thus even if there are no more reflections at the boundary there will still be reflections at the edge of the absorbing layer. The second problem arises due to the angle of incidence. To solve the first one the absorption coefficient can be gradually increased from zero in the simulation domain to the desired maximum damping coefficient in the material. This ramping up of the damping coefficient in the material is given by

$$\kappa = \kappa_{max} \left(\frac{x}{d}\right)^5 \quad (6)$$

The second problem is solved through splitting the pressure in to an orthogonal and a parallel component. This doesn't necessarily make sense for a scalar but this can simply be thought of as orthogonal and parallel contributions to the pressure caused by the orthogonal and parallel components of the velocity. This can be seen in eq ??

$$\frac{\partial o_x}{\partial t} + \frac{\partial p_x}{\partial x} = 0 \quad (7)$$

$$\frac{\partial o_y}{\partial t} + \frac{\partial p_y}{\partial y} = 0 \quad (8)$$

$$\frac{\partial p_x}{\partial t} + \frac{\partial o_x}{\partial x} = 0 \quad (9)$$

$$\frac{\partial p_y}{\partial t} + \frac{\partial o_y}{\partial y} = 0 \quad (10)$$

where the assumption has been made that the all 4 boundaries are either parallel with with the y-axis or x-axis(i.e a rectangle).

The only thing still to be added to these equations is a damping coefficient. To get the desired result and full damping of the wave the damping coefficients for the orthogonal components should be equal to each-other while the parallel components should be zero and this for each boundary. This was accomplished by solving the four equations with a damping coefficient in the whole of the domain. Where the vertical damping coefficients for the vertical boundaries is zero while the horizontal coefficients are equal to each-other and ramp up from zero in the domain to a κ_{max} in the boundary layer. The exact opposite was done for the horizontal boundaries. This leads to the following equations to be solved:

$$\frac{\partial o_x}{\partial t} + \frac{\partial p_x}{\partial x} + \kappa_x o_x = 0 \quad (11)$$

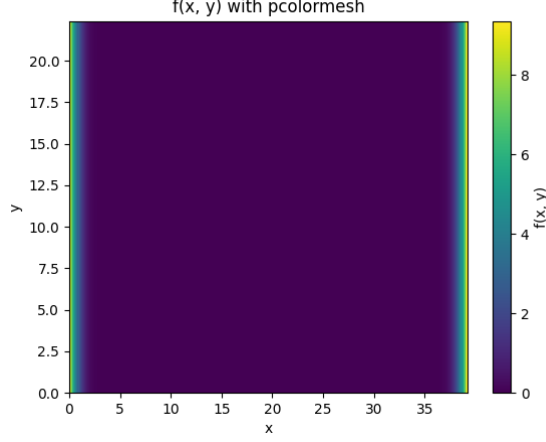


Figure 1: Vertical damping coefficient over the domain

$$\frac{\partial o_y}{\partial t} + \frac{\partial p_y}{\partial y} + \kappa_y o_y = 0 \quad (12)$$

$$\frac{\partial p_x}{\partial t} + \frac{\partial o_x}{\partial x} + \kappa_x p_x = 0 \quad (13)$$

$$\frac{\partial p_y}{\partial t} + \frac{\partial o_y}{\partial y} + \kappa_y p_y = 0 \quad (14)$$

with the following damping coefficients throughout space.

$$\kappa_x = 10 \left(\left(\frac{x - (a - d)}{d} \right)^5 H(x - (a - d)) - \left(\frac{x - d}{d} \right)^5 H(-x + d) \right) \quad (15)$$

$$\kappa_y = 10 \left(\left(\frac{y - (b - d)}{d} \right)^5 H(y - (b - d)) - \left(\frac{y - d}{d} \right)^5 H(-y + d) \right) \quad (16)$$

With a and b the width and height of the domain respectively and H the Heaviside function. This can be seen in the heatmap in figure 1.

2.2 The source

The equations described above will not produce any waves on their own, for this a source is necessary. In these simulations the domain of interest lies for in $kd \in [0.1, 10]$ or thus $f \in [0.0028, 0.28]$. To this end the following source is used.

$$P_{source} = \sin(2\pi f_o) \exp\left(-\frac{(t - t_0)^2}{\sigma^2}\right) \quad (17)$$

This source can be seen in the time domain in fig ?? and in the frequency domain in fig 3. It is clear that the source forms a short band around f_0 spanning the values of kd .

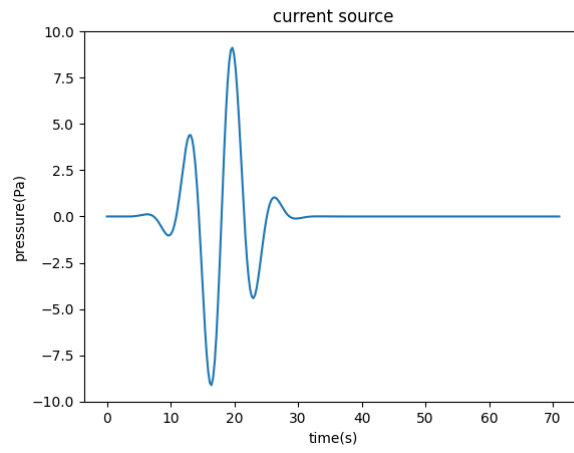


Figure 2: The source in time domain

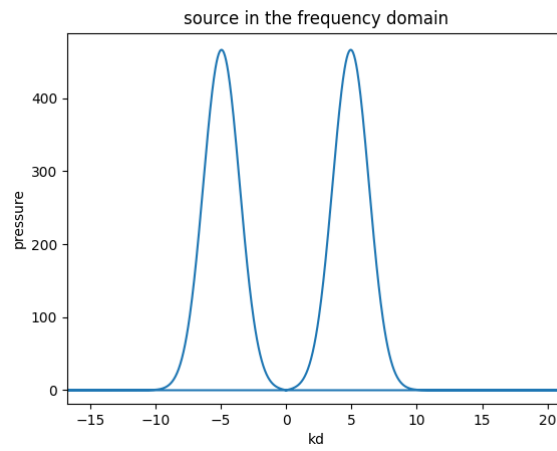


Figure 3: The source in frequency domain

2.3 finite difference equations

Now to solve these equations the pressure and velocity fields need to be discretised. This sampling can be done the same for all fields i.e evaluating all the fields in the same points(a collocated grid)

$$p_x = p_x|_{i,j}^l = p_x(idx, jdy, ldt) \quad p_y = p_y|_{i,j}^l = p_y(idx, jdy, ldt) \quad (18)$$

$$o_x = o_x|_{i+,j}^l = o_x(idx, jdy, ldt) \quad o_y = o_y|_{i,j}^l = o_y(idx, jdy, ldt) \quad (19)$$

but as will be further explained later, this is a bad idea because it leaves half the points untouched for pressure and the other half untouched for the velocities. This is why a staggered grid was chosen so that the fields are evaluated as seen in

$$p_x = p_x|_{i,j}^l \quad p_y = p_y|_{i,j}^l \quad o_x = o_x|_{i+1/2,j}^{l+1/2} \quad o_y = o_y|_{i,j+1/2}^{l+1/2} \quad (20)$$

Now to solve for these discretised functions the equations also need to be discretised. Seeing as the derivatives are based on continuous functions. This can be done through central finite difference based on the Taylor expansion.

$$f(x + dx/2) = f(x) + f'(x)\frac{dx}{2} + \frac{1}{2}f''(x)\frac{dx^2}{4} + \dots \quad (21)$$

$$f(x - dx/2) = f(x) - f'(x)\frac{dx}{2} + \frac{1}{2}f''(x)\frac{dx^2}{4} + \dots \quad (22)$$

Subtracting these two equations and rearranging gives

$$f'(x) = \frac{f(x + dx/2) - f(x - dx/2)}{dx} \quad (23)$$

Where it is clear that this approximation is second order accurate seeing as only third order terms and higher are omitted. For derivatives in space this becomes

$$\frac{\partial p_x}{\partial x}|_{i+1/2,j}^l = \frac{p_x|_{i+1,j}^l - p_x|_{i,j}^l}{dx} \quad \frac{\partial p_y}{\partial y}|_{i,j+1/2}^l = \frac{p_y|_{i,j+1}^l - p_y|_{i,j}^l}{dy} \quad (24)$$

$$\frac{\partial o_x}{\partial x}|_{i,j}^{l+1/2} = \frac{o_x|_{i+1/2,j}^{l+1/2} - o_x|_{i-1/2,j}^{l+1/2}}{dx} \quad \frac{\partial o_y}{\partial y}|_{i,j}^{l+1/2} = \frac{o_y|_{i,j+1/2}^{l+1/2} - o_y|_{i,j-1/2}^{l+1/2}}{dy} \quad (25)$$

and for derivatives in time this becomes

$$\frac{\partial p_x}{\partial t}|_{i,j}^{l+1/2} = \frac{p_x|_{i,j}^{l+1} - p_x|_{i,j}^l}{dt} \quad \frac{\partial p_y}{\partial t}|_{i,j}^{l+1/2} = \frac{p_y|_{i,j}^{l+1} - p_y|_{i,j}^l}{dt} \quad (26)$$

$$\frac{\partial o_x}{\partial t}|_{i+1/2,j}^l = \frac{o_x|_{i+1/2,j}^{l+1/2} - o_x|_{i+1/2,j}^{l-1/2}}{dt} \quad \frac{\partial o_y}{\partial t}|_{i,j+1/2}^l = \frac{o_y|_{i,j+1/2}^{l+1/2} - o_y|_{i,j+1/2}^{l-1/2}}{dt} \quad (27)$$

When substituting this in the equation the following equations can be found

$$o_x|_{i+1/2,j}^{l+1/2} = o_x|_{i+1/2,j}^{l-1/2} - \frac{dt}{dx}(p_x|_{i+1,j}^l - p_x|_{i,j}^l) - \kappa_x o_x|_{i+1/2,j}^l \quad (28)$$

$$o_y|_{i,j+1/2}^{l+1/2} = o_y|_{i,j+1/2}^{l-1/2} - \frac{dt}{dx}(p_x|_{i,j+1}^l - p_x|_{i,j}^l) - \kappa_y o_y|_{i,j+1/2}^l \quad (29)$$

$$p_x|_{i,j}^{l+1} = p_x|_{i,j}^l - \frac{dt}{dx}(p_x|_{i+1/2,j}^{l+1/2} - o_x|_{i+1/2,j}^{l-1/2}) - \kappa_x p_x|_{i,j}^{l+1/2} \quad (30)$$

$$p_y|_{i,j}^{l+1} = p_y|_{i,j}^{l-1} - \frac{dt}{dx}(o_y|_{i,j+1/2}^{l+1/2} - o_y|_{i,j+1/2}^{l-1/2}) - \kappa_y p_y|_{i,j}^{l+1/2} \quad (31)$$

Where each next time step can be calculated based on the previous step. There is still a problem, however, seeing as for the damping one needs to know the value of o_x and o_y at time steps l and p_x and p_y at time steps $(l+1/2)dt$ even-though these values are not evaluated at those points in the grid. To solve this issue interpolation is used as can be seen in eq.

$$\kappa_x o_x|_{i+1/2,j}^l = \kappa_x \frac{o_x|_{i+1/2,j}^{l-1/2} + o_x|_{i+1/2,j}^{l+1/2}}{2} \quad (32)$$

$$\kappa_y o_y|_{i,j+1/2}^l = \kappa_y \frac{o_y|_{i,j+1/2}^{l-1/2} + o_y|_{i,j+1/2}^{l+1/2}}{2} \quad (33)$$

$$\kappa_x p_x|_{i+1/2,j}^l = \kappa_x \frac{p_x|_{i,j}^l + p_x|_{i,j}^{l+1}}{2} \quad (34)$$

$$\kappa_y p_y|_{i+1/2,j}^l = \kappa_y \frac{p_y|_{i,j}^l + p_y|_{i,j}^{l+1}}{2} \quad (35)$$

Now each next time step can be calculated based on the previous step. This means only an initial condition for the field in each point is required and the field through out time and space can be calculated. This initial value is chosen as zero in each point. However with these equations the field would remain zero seeing as a source is still to be added. To do this the value of the source at each time step is divided by two and added to both pressure equations(at the position of the source of-course). This way the total pressure at each time is still the pressure of the source and this total pressure is equally divided between the two contributions. After including the source and interpolation these are the final equations

$$(1 + \kappa_x \frac{dt}{2}) o_x|_{i+1/2,j}^{l+1/2} = (1 - \kappa_x \frac{dt}{2}) o_x|_{i+1/2,j}^{l-1/2} - \frac{dt}{dx}(p_x|_{i+1,j}^l - p_x|_{i,j}^l) \quad (36)$$

$$(1 + \kappa_y \frac{dt}{2}) o_y|_{i,j+1/2}^{l+1/2} = (1 - \kappa_y \frac{dt}{2}) o_y|_{i,j+1/2}^{l-1/2} - \frac{dt}{dx} (p_x|_{i,j+1}^l - p_x|_{i,j}^l) \quad (37)$$

$$(1 + \kappa_x \frac{dt}{2}) p_x|_{i,j}^{l+1} = (1 - \kappa_x \frac{dt}{2}) p_x|_{i,j}^l - \frac{dt}{dx} (p_x|_{i+1/2,j}^{l+1/2} - o_x|_{i+1/2,j}^{l-1/2}) \quad (38)$$

$$(1 + \kappa_y \frac{dt}{2}) p_y|_{i,j}^{l+1} = (1 - \kappa_y \frac{dt}{2}) p_y|_{i,j}^{l-1} - \frac{dt}{dx} (o_y|_{i,j+1/2}^{l+1/2} - o_y|_{i,j+1/2}^{l-1/2}) \quad (39)$$

2.4 stability and accuracy

To make sure the solution is stable the time step is chosen in such a way that combined with the spatial steps the Courant condition is fulfilled.

$$CN \leq 1 \quad (40)$$

with CN the Courant number

$$CN = \sqrt{c^2 dt^2 (\frac{1}{dx^2} + \frac{1}{dy^2})} \quad (41)$$

On the other hand the dt must be as big as possible for accuracy and so dt is chosen such that

$$CN = 1 \quad (42)$$

2.5 finite difference boundary conditions

2.5.1 empty

With the equations described in the section above the simulation of an empty domain can now be calculated. However boundary conditions on the edge of the simulation domain still need to be implemented seeing as the domain is still finite. This boundary condition was chosen as perfectly hard. This is implemented by leaving all the values at the edges at zero.

2.5.2 Infinitely thin structure

For the infinity thin structure the parameters of the empty simulation can be reused but this time a perfectly hard floor and wall need to be included. This can be done by assuming the plane representing the floor is at a plane where o_y is discretised and to put o_y on this plane equal to zero at each time step. To include the wall, the assumption is made that it lies in the plane where o_x is discretised. Then the values of o_x between the ground and the height of the wall in this plane are put equal to zero at each step. This can be seen in the following formulas

$$o_y|_{i,J+1/2}^{l+1/2} = 0 \quad (43)$$

$$o_x|_{I+1/2,j}^{l+1/2} = 0 \quad (44)$$

with $J+1/2$ the index of the ground plane, $I+1/2$ the index of the wall plane, j the indexes between J and the index at the top of the wall and i spanning all indexes in the domain.

2.5.3 Thick object

The procedure to include a hard wall and a hard thick object are completely analogous to the the infinitely thin structure. The only difference is that now there are two walls. These are implemented in the exact same way as the one wall in the previous section. And lastly a "ceiling" needs to be implemented which is done in the same way as the floor but at a different height and instead of spanning the entire horizontal space, only the x values between the walls is considered.

$$o_y|_{i,J_{ground}+1/2}^{l+1/2} = 0 \quad (45)$$

$$o_x|_{I1+1/2,j}^{l+1/2} = 0 \quad (46)$$

$$o_x|_{I2+1/2,j}^{l+1/2} = 0 \quad (47)$$

$$o_y|_{ind,J_{ceiling}+1/2}^{l+1/2} = 0 \quad (48)$$

with $J_{ground}+1/2$ the index of the ground plane, $I1+1/2$ and $I2+1/2$ the index of the left wall- and right wall plane respectively, $J_{ceiling} + 1/2$ the index of the ceiling plane, ind the indexes spanning from the left wall to the right wall and j the indexes going from the ground to the ceiling.

To implement a real impedance one needs to take a look at the update equations on the boundary for example on the left wall it looks like

$$o_x|_{I1+1/2,j}^{l+1/2} = o_x|_{I1+1/2,j}^{l-1/2} - \frac{2dt}{dx}(p|_{I1+1/2,j}^l - p|_{I1,j}^l) \quad (49)$$

Where a one sided finite difference was used seeing as the pressure beyond the boundary doesn't exist. Using the real impedance relation

$$p|_{I1+1/2,j}^l = Z o_x|_{I1+1/2,j}^l \quad (50)$$

the pressure on the boundary can be found in terms of the velocity on the boundary which is ideal seeing as it is a boundary where o_x is being discretised. However it is not ideal that the velocity needs to be evaluated at a time step ldt . This problem can be solved through interpolation as follows

$$o_x|_{I1+1/2,j}^l = \frac{o_x|_{I1+1/2,j}^{l+1/2} + o_x|_{I1+1/2,j}^{l-1/2}}{2} \quad (51)$$

after substituting this in and rearranging the following formula is achieved

$$(1 + Z \frac{dt}{dx}) o_x|_{I1+1/2,j}^{l+1/2} = (1 - Z \frac{dt}{dx}) o_x|_{I1+1/2,j}^{l-1/2} + \frac{2dt}{dx} p|_{I1,j}^l \quad (52)$$

and analogously for the other boundaries

$$(1 + Z \frac{dt}{dx}) o_x|_{I2+1/2,j}^{l+1/2} = (1 - Z \frac{dt}{dx}) o_x|_{I2+1/2,j}^{l-1/2} - \frac{2dt}{dx} p|_{I2,j}^l \quad (53)$$

$$(1 + Z \frac{dt}{dx}) o_y|_{ind, J_{ceiling}-1/2}^{l+1/2} = (1 - Z \frac{dt}{dx}) o_y|_{ind, J_{ceiling}-1/2}^{l-1/2} - \frac{2dt}{dx} p|_{J_{ceiling}, j}^l \quad (54)$$

Where the opposite one sided difference is used in combination with the following impedance relationships

$$p = -Z o_x \quad (55)$$

$$p = -Z o_y \quad (56)$$

seeing as the direction of the impinging velocity is now against the unit axes.

2.5.4 Tilted boundaries

To approach tilted perfectly hard boundary condition a first attempt was made by trying to approximate it with a "stair case" The boundary conditions in this case look like

$$o_y|_{index, 4index-1/2+J_{ground}}^{l+1/2} = 0 \quad (57)$$

$$o_x|_{index+1/2, 4index+J_{ground}}^{l+1/2} = 0 \quad (58)$$

$$o_x|_{index+1/2, 4index+1+J_{ground}}^{l+1/2} = 0 \quad (59)$$

$$o_x|_{index+1/2, 4index+2+J_{ground}}^{l+1/2} = 0 \quad (60)$$

$$o_x|_{index+1/2, 4index+3+J_{ground}}^{l+1/2} = 0 \quad (61)$$

for the positive slope and

$$o_y|_{index+middle+I1+1, 4middle+3-4index-1/2}^{l+1/2} = 0 \quad (62)$$

$$o_x|_{index+middle+I1+1+1/2, 4middle+3-4index}^{l+1/2} = 0 \quad (63)$$

$$o_x|_{index+middle+I1+1+1/2, 4middle+3-4index-1}^{l+1/2} = 0 \quad (64)$$

$$o_x|_{index+middle+I1+1+1/2, 4middle+3-4index-2}^{l+1/2} = 0 \quad (65)$$

$$o_x|_{index+middle+I1+1+1/2, 4middle+3-4index-3}^{l+1/2} = 0 \quad (66)$$

for the negative slope. Where *index* denotes all the indexes between zero and *middle*, the amount of points between the start of the obstacle and the middle. In this approximation o_x represents "vertical parts" of each staircase step and o_y the horizontal part of every staircase step. (It is important to note that the vertical coordinate only start from the floor index so for each vertical index there should be a +floor index). A real impedance can now be imposed by using the same formula's from thick wall. For the horizontal parts(o_y) the same formula as for the ceiling of the thick wall can be used while for the vertical parts(o_x) of the staircase the same formula as the walls of the tick wall can be used.

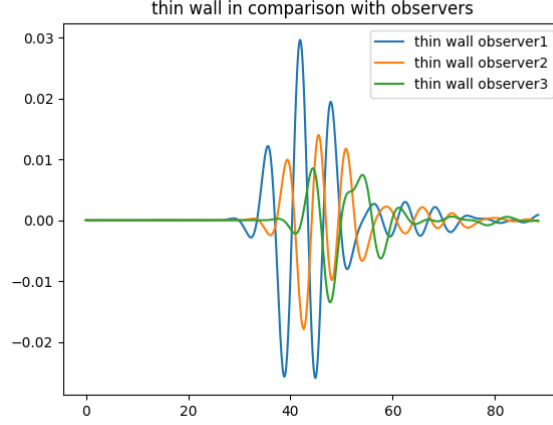


Figure 4: Thin wall as measured by observers throughout time

3 Infinitely Thin Structure

The analytical solution for the infinitely thin wall can be calculated by using the formula from the course.

This is however not the full contribution. Seeing as this is meant for a source and an obstacle in free space while our set up has a ground plane. This plane can be removed if another virtual source and wedge is added at the perfect reflected position (around the ground plane). This introduces 2 wedges and 2 sources and 4 contributions. When adding all these up one gets the analytical solution.

3.1 Results

In figure 4 the field, as measure by the different observers, is given in the time domain. When comparing the source with the observations it is clear that there is a time delay that becomes bigger the further away from the source one gets. This makes sense as the wave needs to travel further away. Additionally there seems to be a decrease in amplitude that also scales with distance. This also makes sense seeing that the wave radiates from the source in all directions spreading the energy. Thus the further away from the source the more spread out the energy will be and lower the amplitude will be. The tail doesn't seem as smooth as would be expected this could be due to a reflecting wave coming from the ground.

When comparing the observation for empty space and thin wall it is clear that there is also a delay and decrease in amplitude caused by the fact that the wave can no longer travel directly to the observation point.

In figure 5 the relative field w.r.t. the free field is plotted in the frequency domain at the locations of the observers. In figure 6 the analytical solution is

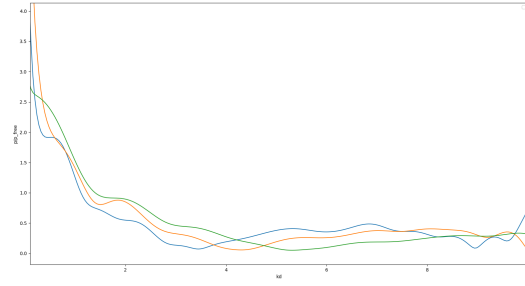


Figure 5: Relative field with thin wall in frequency domain

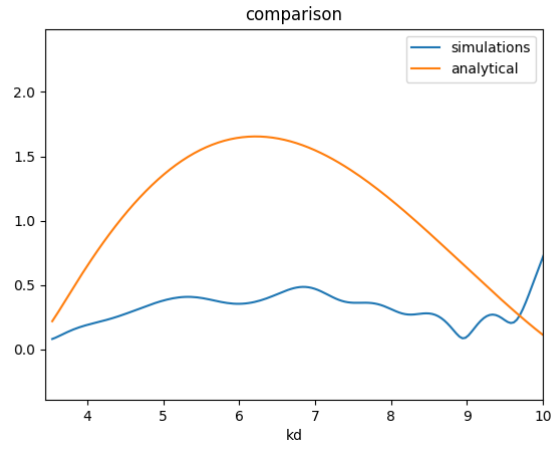


Figure 6: Relative field with thin wall numerical and analytical in frequency domain

compared to the numerical one, at the first observer's position. With some goodwill the overall shape from the analytical solution can be seen in the numerical one, including some bumps.

4 Thick Object with Finite Impedance

In figure 7 a thick wall which is perfectly hard is compared to the infinitely thin wall in time domain. When considering the thick wall the effect seen in the comparison between empty space and the thin wall is larger here but not by much. Seeing as the wall now has a thickness it takes a longer time for the wave to get past the obstacle and the peak around which the wave diffracts is also less sharp. This means that the wave will arrive later and that less of the total energy arrives. When including a real impedance (figure 8) some absorption can happen meaning way less of the energy makes it past the obstacle. Seeing as the impedance $2c$ is chosen the impedance lies between c and infinity meaning the wave will partially reflect (something) and partially absorb which is observed in the simulation.

In figures 9 and 10 the relative field w.r.t. the free field is plotted in the frequency domain at the locations of the observers for a hard rectangle and one with a finite impedance respectively. For higher frequencies both match well. For lower frequencies the finite impedance seems to not notice the obstacle, while the hard rectangle has some nulls in the lower frequency-range.

5 Tilted Boundaries

When including a perfectly hard object with tilted boundaries (figure 11). It seems like it does still take more time to arrive at the observation point relative to a perfectly hard wall. However the amplitude seems higher. This might be due to the fact that the wave diffracts easier around the sharp peak allowing for more energy to get passed the object. When a real impedance is included (figure 12) however the amplitude decreases significantly. This is only logical seeing as a lot of energy will now be absorbed by the obstacle.

In figures 13 and 14 the relative field w.r.t. the free field is plotted in the frequency domain at the locations of the observers. The analysis is practically the same as for the rectangle. For higher frequencies they match, for lower not so much.

5.1 Refinement

In order to better model the non-cartesian surface, we included a grid refinement around the tilted boundaries.

This was achieved by making dx and dy functions of respectively x and y . As a first attempt both were made a step-function, in which around the boundaries they are both half of what they are elsewhere. This however results in reflections at the discontinuity in dx or dy , which can be seen when looking

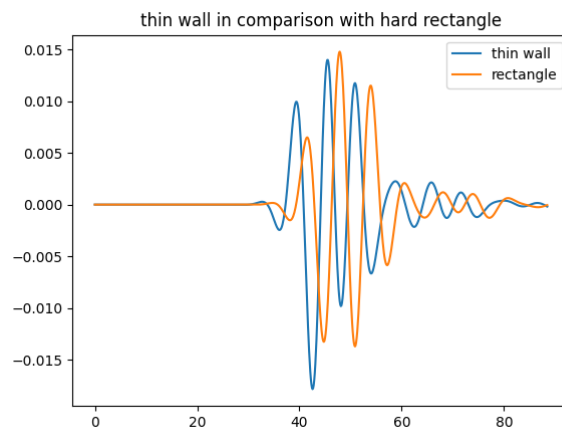


Figure 7: Hard thick wall and thin wall in time domain

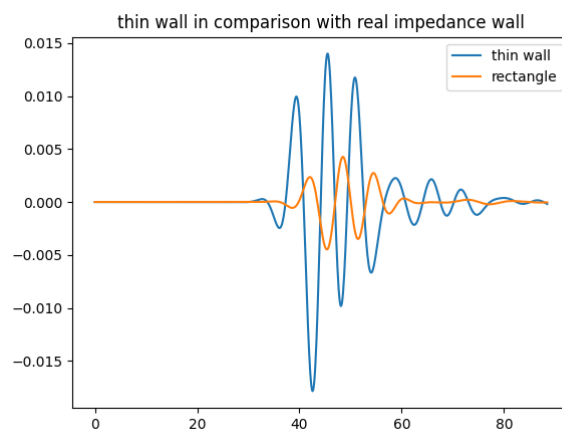


Figure 8: Thick wall with finite impedance and thin wall in time domain

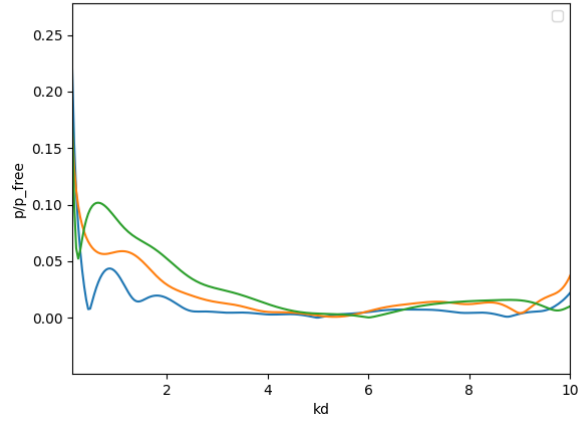


Figure 9: Relative field with hard rectangle in frequency domain

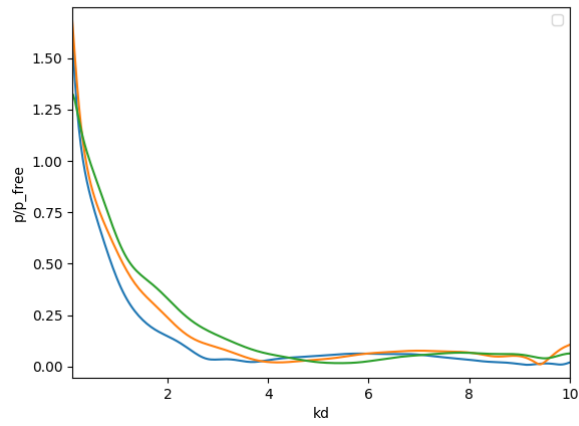


Figure 10: Relative field with rectangle with finite impedance in frequency domain

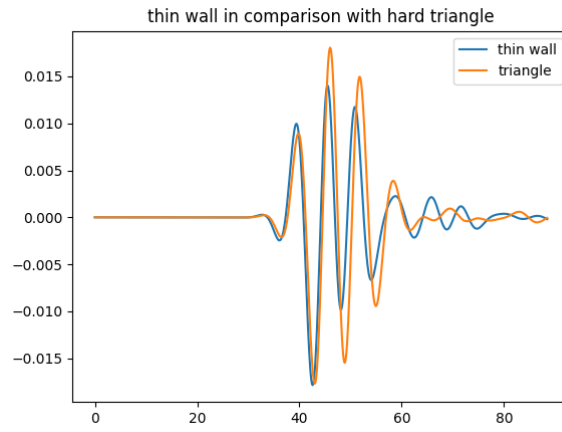


Figure 11: Hard tilted boundaries compared to thin wall in time domain

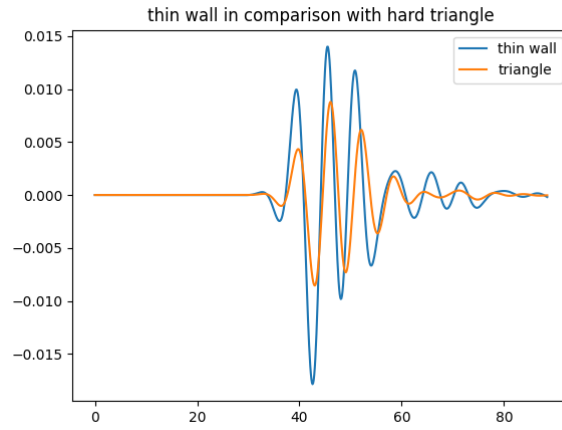


Figure 12: Tilted boundaries with finite impedance compared to thin wall in time domain

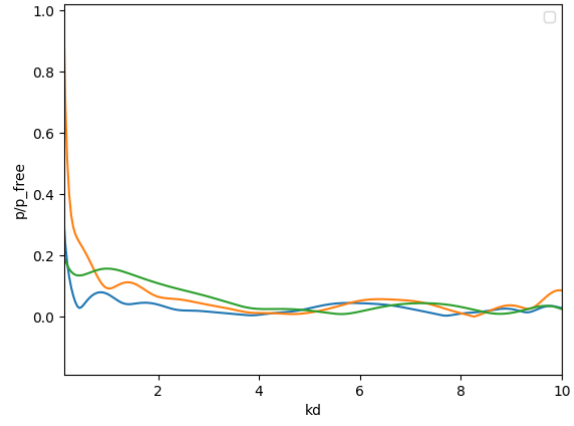


Figure 13: Relative field with hard wedge in frequency domain

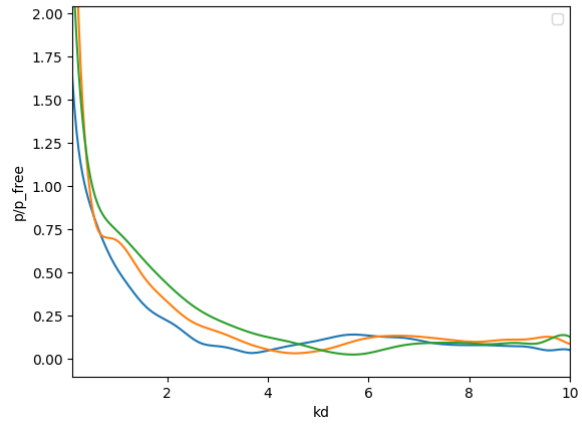


Figure 14: Relative field with wedge with finite impedance in frequency domain

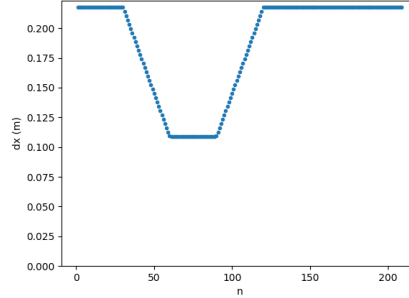


Figure 15: dx after refinement (linear)

at an animation of a short Gaussian pulse. Next, to remove these reflections, a buffer zone was added in both x - and y -direction in which dx and dy are linearly interpolated (see fig 15).

Both these were in practice implemented by taking the original discretisation coordinates, and transforming these, either parts-wise linearly (for the step-function in dx or dy), or with a parabolic part in the buffer-zone (for the linear dx and dy).

There were dx or dy is not constant however, this results in an error in the update scheme. As, for example, the discretised o_x may no longer be halfway between two p 's. This means that in the update equations of o_x , it is erroneous to use $\frac{dp}{dx}$ as calculated by differentiating two neighbouring p 's. So an interpolation was made between three values of $\frac{dp}{dx}$ at the halfway point between two p 's: two closest neighbours to the left, the two direct neighbours, and those to the right. Since there are three values, a parabolic interpolation was optimal. At the edges, where there are only two values to interpolate between, a linear interpolation was used instead. To update o_x , the interpolated $\frac{dp}{dx}$ at the location of o_x (after the coordinate transformation) was used.

Figure 17 gives the relative field in the frequency domain for a perfectly hard wedge. The observers, from closest to furthest from the wedge are, blue, orange and green. As a counterpart in figure 16 gives the same problem without refinement. It is quite clear that something went wrong here, the relative field shows strong peaks for two of the observers when including refinement. A possibility is the method used for the boundary. It has a tendency to create oscillations which stay long after the wave has passed. This might explain the peaks visible, however, the figure without refinement uses the same method.

The code for a finite Z could not easily be converted to work with the refinement method used. This combined with a shortage of time means only a perfectly hard boundary was studied.

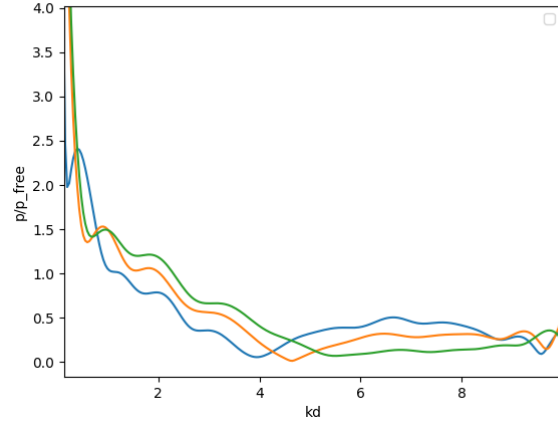


Figure 16: Relative field without refinement for wedge with $Z = 0$

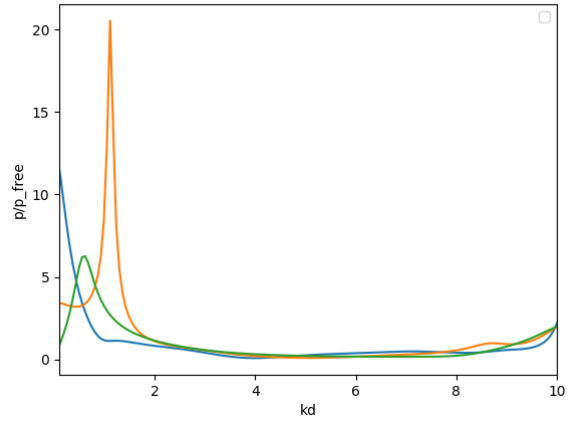


Figure 17: Relative field with refinement for wedge with $Z = 0$

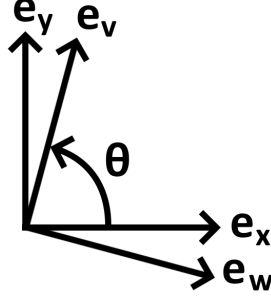


Figure 18: Important unit vectors

5.2 Non Cartesian Grid

Another route that was explored is to tilt the grid. In this way the surface of the tilted boundary becomes flat on one side, thus removing the problems that arise from a staircase-like boundary. The left-side was chosen to be flat, as the most important reflections occur there.

Instead of working with x - and y -coordinates, a switch is made to u - and v -coordinates. These are related via equations 67 and 68, in which θ is the angle between the v -axis and the x -axis.

$$x = u + v \cos \theta \quad (67)$$

$$y = v \sin \theta \quad (68)$$

In figure 18 the basis-vectors are given. Note that $\vec{e}_x = \vec{e}_u$, but $x \neq u$. In this figure another vector, $\vec{e}_w$, is introduced. This vector is perpendicular to $\vec{e}_v$. With this figure the relations given in equations 69 and 70 are easily derived.

$$\vec{e}_w = \sin \theta \vec{e}_x - \cos \theta \vec{e}_y \quad (69)$$

$$\vec{e}_v = \cos \theta \vec{e}_x + \sin \theta \vec{e}_y \quad (70)$$

There is still an important choice remaining, namely, which directions of $\vec{o}$ to store. A first idea might be to choose $o_u = \vec{e}_u \cdot \vec{o}$ and $o_v = \vec{e}_v \cdot \vec{o}$, as analogous to the cartesian grid. Remember however that the goal is to implement a tilted perfectly hard boundary, which has as boundary condition that $o_n = 0$, with n the normal of the boundary. For this boundary condition to be implemented simply, o_w and o_v are chosen, in which $\vec{e}_w$ is the normal of the tilted boundary (on the left side). An added benefit is that now both stored directions of $\vec{o}$ are orthogonal to eachother, so no data overlaps. Figure 19 shows the location of a few p , o_w and o_v 's.

5.2.1 Update Scheme

Now the update scheme can be adjusted for the tilted grid. To update p , the divergence of $\vec{o}$ is needed at the location of p , with $\nabla \cdot \vec{o} = \frac{do_w}{dw} + \frac{do_v}{dv}$. The

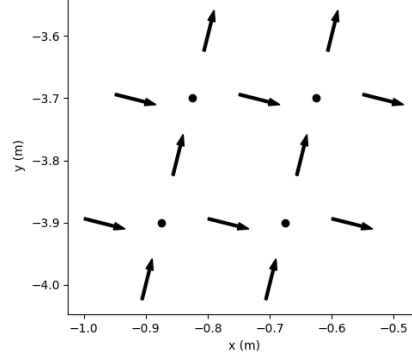


Figure 19: Discretised fields

following quantities are known:

$$\frac{do_v}{dv}|_{v_0+\Delta v/2} \cdot \Delta v \approx o_v|_{v_0+\Delta v} - o_v|_{v_0},$$

$$\frac{do_w}{du}|_{u_0+\Delta u/2} \cdot \Delta u \approx o_w|_{u_0+\Delta u} - o_w|_{u_0}.$$

$\frac{do_w}{dw}$ requires some more attention. From equations 70 and 69, the following can be derived:

$$\vec{e}_w = \frac{1}{\sin \theta} (\vec{e}_u - \cos \theta \vec{e}_v).$$

From which

$$\frac{do_w}{dw} = \vec{e}_w \cdot \nabla o_w = \frac{1}{\sin \theta} \left(\frac{do_w}{du} - \cos \theta \frac{do_w}{dv} \right)$$

follows. So the missing piece is $\frac{do_w}{dv}$, which can only be interpolated at p 's location by using

$$\frac{do_w}{dv}|_{v_0} \cdot 2\Delta v \approx o_w|_{v_0+\Delta v} - o_w|_{v_0-\Delta v},$$

combined with

$$o_w|_{u_0+\Delta u/2} \approx \frac{o_w|_{u_0} + o_w|_{u_0+\Delta u}}{2}.$$

For the update equations of $\vec{o}$ an analogous approach is followed.

In a hand-waving approach a rough approximation was made for the Courant Number. Starting from the standard form, and substituting $dx = du + dv \cos \theta$ and $dy = dv \sin \theta$ (derived from equations 67 and 68), equation 71 is calculated.

$$\text{CN} = c^2 dt^2 ((du + dv \cos \theta)^{-2} + (dv \sin \theta)^{-2}) \quad (71)$$

For stability to be achieved, $\text{CN} \approx 0.8$ is the maximum (for $\tan \theta = 4$). The adjusted update scheme involves a lot of interpolation of missing values, which decrease the stability.

5.2.2 Perfectly Matched Layers

$$\frac{dp_{\perp}}{dt} + c^2 \frac{do_{\alpha}}{d\alpha} + \kappa_{\perp} p_{\perp} = 0 \quad (72)$$

$$\frac{dp_{\parallel}}{dt} + c^2 \frac{do_{\gamma}}{d\gamma} + \kappa_{\parallel} p_{\parallel} = 0 \quad (73)$$

The perfectly matched layers in the tilted grid become a bit more complex. This especially for the upper PML. The update equations for $p = p_{\perp} + p_{\parallel}$ are given by equations 72 and 73. Looking at the upper PML, so that the perpendicular direction is $\vec{e}_y$ and the parallel one is $\vec{e}_x$, these equations become:

$$\frac{dp_{\perp}}{dt} + c^2 \frac{do_y}{dy} + \kappa_{\perp} p_{\perp} = 0,$$

$$\frac{dp_{\parallel}}{dt} + c^2 \frac{do_x}{dx} + \kappa_{\parallel} p_{\parallel} = 0.$$

By defining $\vec{p} = p_{\perp} \vec{e}_y + p_{\parallel} \vec{e}_x$ and $\vec{\kappa} = \kappa_{\perp} \vec{e}_y + \kappa_{\parallel} \vec{e}_x$, these can be rewritten in one equation as follows

$$\frac{d}{dt} \vec{e}_i \cdot \vec{p} + c^2 \vec{e}_i \cdot \nabla \vec{o} \cdot \vec{e}_i + |\vec{e}_i \cdot \vec{\kappa}| \vec{p} \cdot \vec{e}_i = 0$$

The absolute value of $\vec{e}_i \cdot \vec{\kappa}$ is important: if we chose $-\vec{e}_y$ instead of $\vec{e}_y$ as the perpendicular direction, the same equation should be gotten. Intuitively it is also non-sensical to have a negative damping-coefficient, as this would amplify. By now substituting $\vec{e}_i = \vec{e}_w$ or $\vec{e}_i = \vec{e}_v$, the equations for p_w and p_v are received.

It is important to note that this last step is not valid. Since this is an equation involving tensors, it cannot be generalized to any vector starting from two basis-vectors. This method, however, results in easy to implement PML's and, by visual validation, results in correct absorption.

5.2.3 Perfectly Hard Boundaries

The tilted boundary on the left side of the triangle is, as mentioned above, easy to implement. It just requires setting $o_w = 0$. The right-hand side isn't as simple to implement correctly (ie. such that only $o_n = 0$), instead a slightly incorrect alternative was used, setting all components of $\vec{o}$ to zero. Lastly the ground-plane had to be implemented. Neither of the simple approaches above are viable here, as $o_n = o_y$ is not stored directly so can't be set to zero, and neither can all components of $\vec{o}$ be set to zero. This last one is due to the fact that o_w and o_v aren't discretised at the same location, so that in reality the incoming wave will either see $o_w = 0$ or $o_v = 0$.

With this there is only one solution remaining, calculate o_y at the boundary by interpolation and remove this part from the stored o_v . In practice the opposite was done, o_x was calculated and o_v was set to only include this part. Approximating o_w at the ground by the one just above it, and using $o_x = o_w \sin \theta + o_v \cos \theta$, yields o_x . This boundary condition does not prevent

waves from entering the boundary, which was mitigated by setting p equal to zero under the ground.

Finally another approach was tried. The goal is for o_y to be zero at the ground. Looking at its update equation,

$$\frac{do_y}{dt} + \frac{dp}{dy} = 0,$$

and using $o_y = 0$ for all t at $y = 0$, it follows that $\frac{dp}{dy} = 0$ at $y = 0$. As done before, this derivative can be written in terms of derivatives to u and v . The derivative towards v is easily approximated at the ground by using central difference, but towards u requires interpolation. From this equation p just below the ground can be set such that the boundary condition is achieved. Sadly this method turned out to be very unstable and it was not useable.

Boundaries with finite impedance were not studied for the tilted grid.

5.2.4 Results

In figure 20 the relative field w.r.t. the free field in the frequency domain is given, as calculated at the three positions of the observers. When comparing this with the same results from the cartesian grid (figure 16), it at first glance seems a different spectrum. But at closer inspection the general peaks do match. The spectrum no longer explodes for small wavelengths and has a fuller body there. For the middle- to high-end of the spectrum both look about the same. Around $kd = 10$ it can be seen that the cartesian grid starts oscillating, this is not the case in the tilted grid and can be seen as a showcase of it working. As higher frequencies, or lower wavelengths are harder to implement correctly with the staircase-like boundary.

This tilted grid does come with a few drawbacks. Mainly it becomes less stable, leading to smaller timesteps and more phase-error. Besides this it makes the implementation of the PML and ground more complex.

6 Conclusion

An infinitely thin perfectly reflecting boundary was implemented alongside perfectly matched layers to simulate the infinite space, it was compared to the analytical solution, but showed little correspondence. Next a rectangle with finite width was implemented, first perfectly hard, and then with a finite impedance. After this a wedge was implemented. To better accommodate this structure a grid-refinement was used, which did not yield very useful results. As an alternate approach a tilted grid was implemented. The tilted grid makes the implementation of the tilted boundary easier and more accurate. But comes with a few drawbacks.

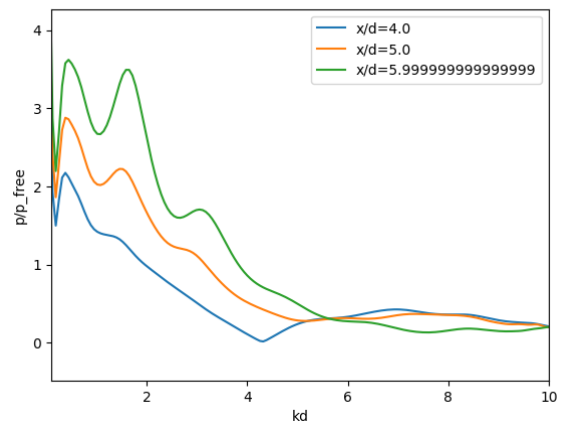


Figure 20: Relative field at three observer points